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Design and Manufacturing

Pantograph Design Report

Overview:

The primary goal of this project is for teams of two to design and construct a pantograph that can accurately and repeatedly magnify an image by a factor of 2.0x. The scope of the project allows teams to decide on the type of materials and marking utensil to be used for the magnified reproduction. This encourages creativity in choosing materials, manufacturing methods, and overall design. This project not only challenges us to apply our theoretical knowledge about tolerances and precision but also requires practical problem-solving to achieve precise mechanical replication.

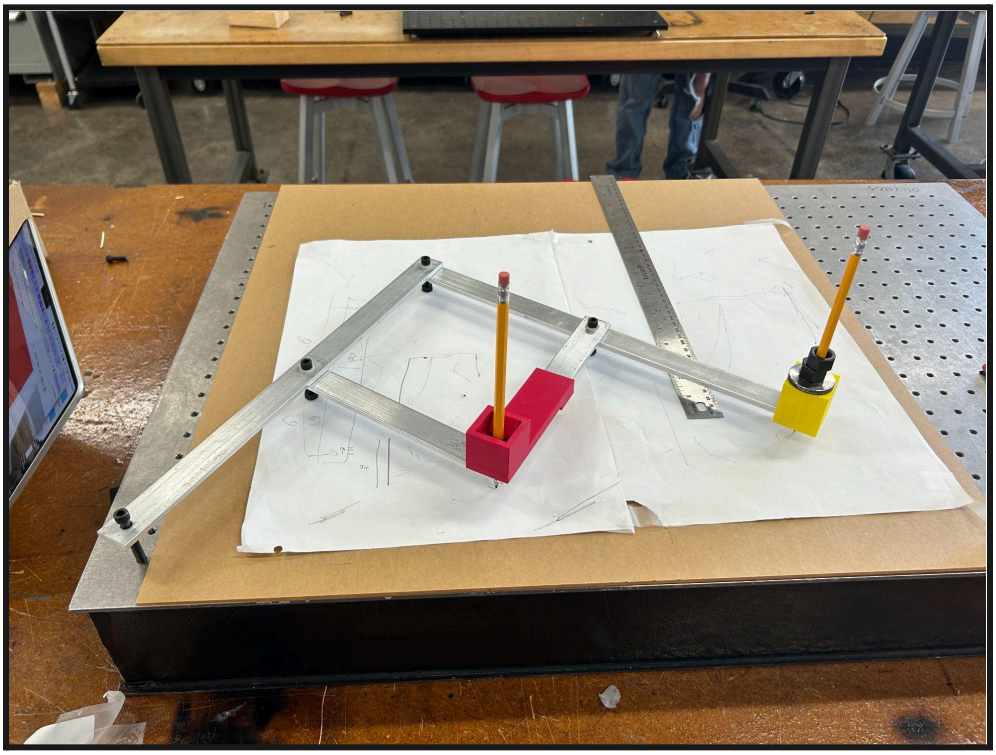
Design Requirements:

- The pantograph must magnify images precisely by 2.0x in all directions and reproduce at least one chosen image accurately.
- It should attach to an optical breadboard plate with specific threading and spacing requirements, allowing for various setup configurations.
- The design should incorporate at least two different types of materials and three distinct manufacturing processes.
- The marking utensil attached to the pantograph must be securely connected but easily replaceable.

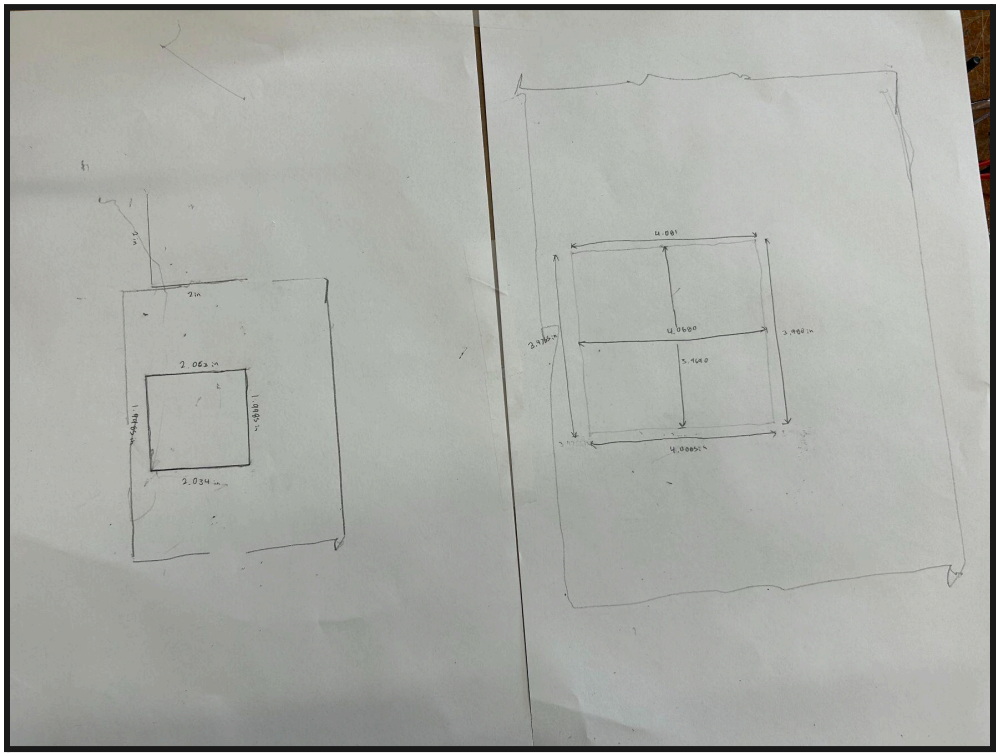
Our Design:

We decided for our machining methods to use the mill, horizontal bandsaw, and 3D-Printers. For our materials we used aluminum bars as well as 3D-Printer plastic filament.

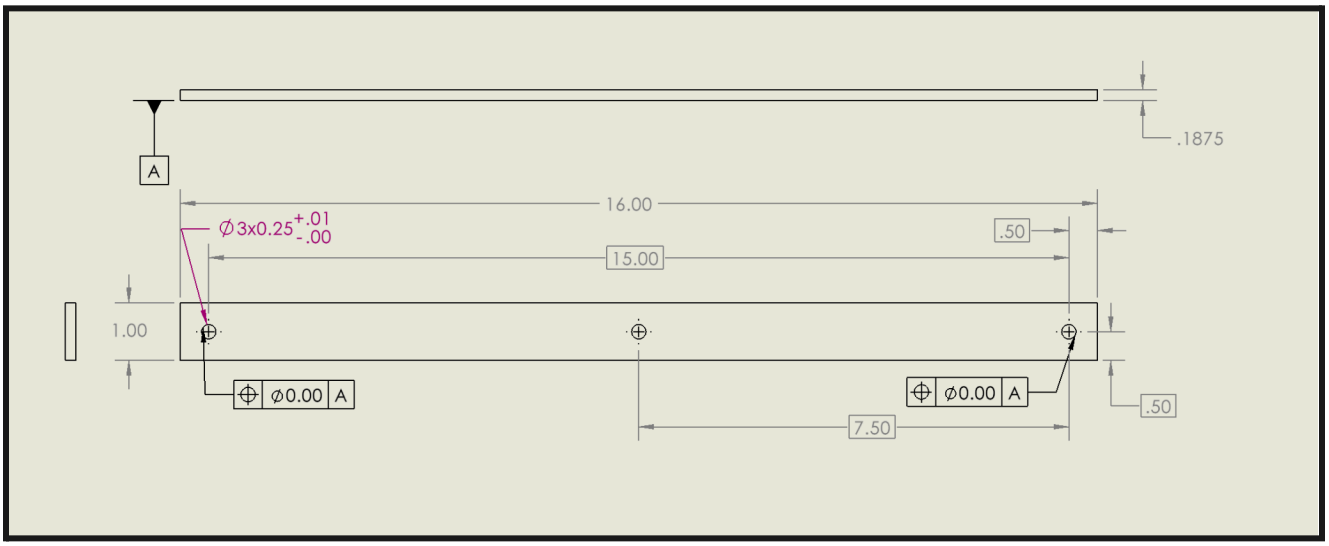
Physical Assembly in Real Life:

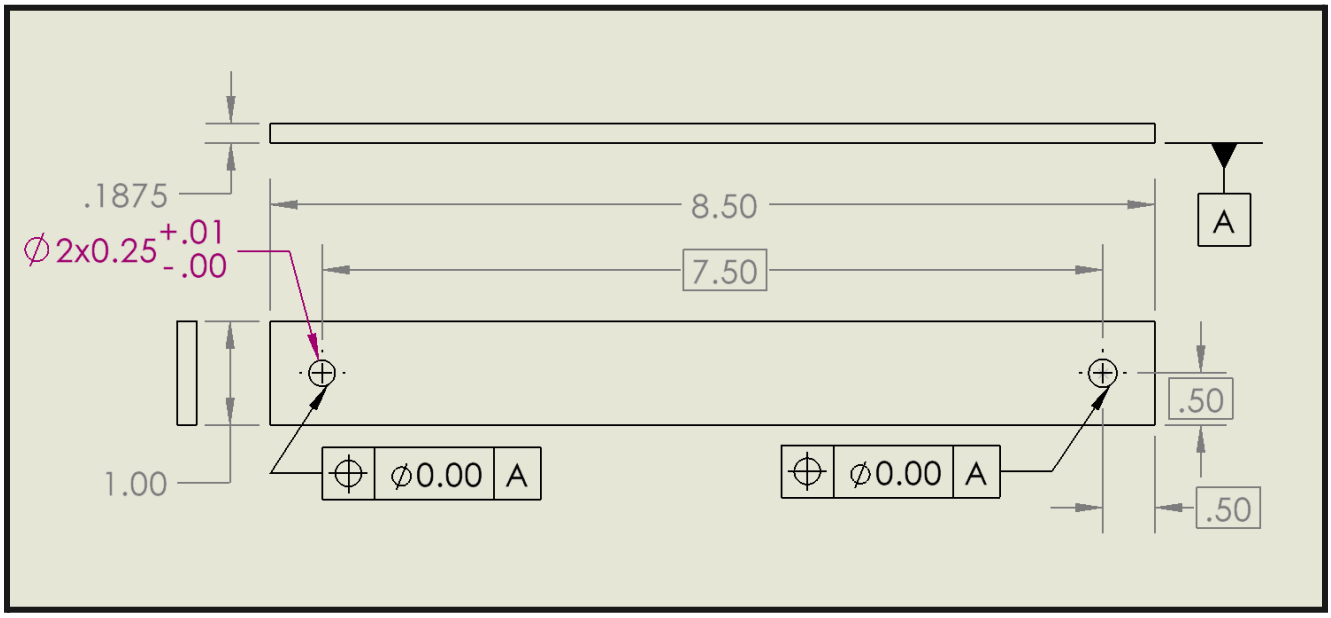


Original and Magnified Images:



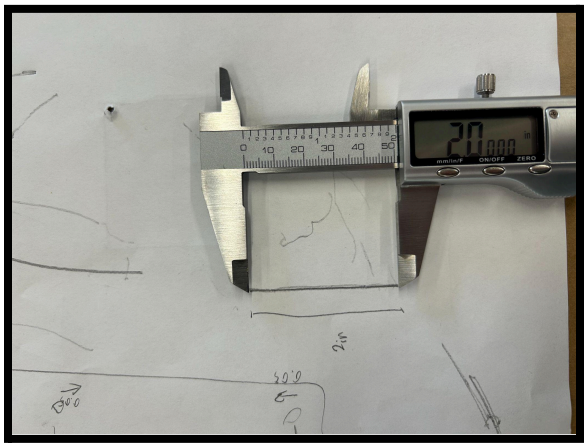
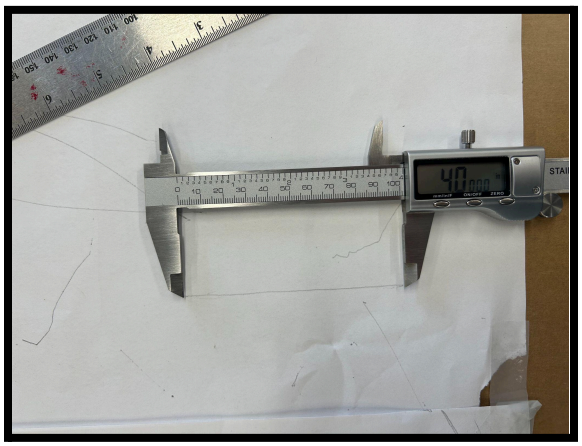
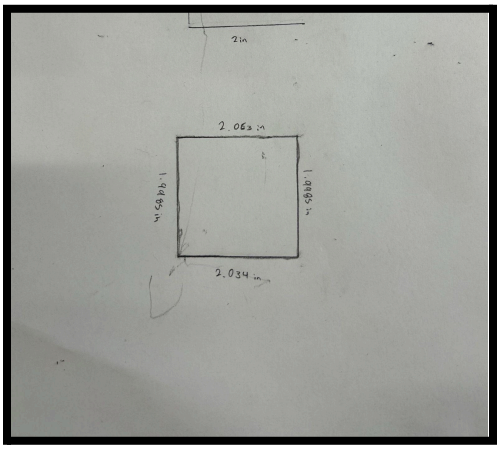
Tolerance Analysis (Updated):

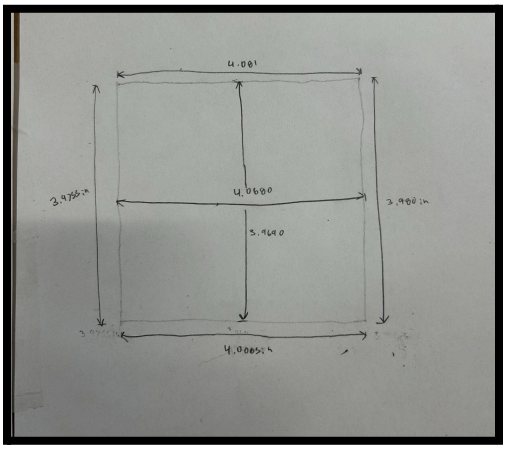




Using ¼-20 bolts as our fastener, we get a Fastener MMC of 0.25in. Then for each hole we are drilling we intend to make it 0.25in in diameter but to account for potential drill wobble or human error we have a tolerance of +0.01in. This gives us a Hole MMC of 0.25in because we look at the smallest size a hole could be so the +0.01in tolerance is not needed. Then doing T=H-F we get T=0.25-0.25=0. If the value of T was not 0, the value would be T for both of the holes as they are floating fasteners. Error in size could occur if the holes are bigger but for true position tolerancing we got 0in.

Calibration Validation:



To complete our calibration validation we did both a 2in line and a 2inx2in box as the original image. For the 2in line we were able to get an exact 2.0x reproduction as shown in the images above. For the 2inx2in box we had some variation in reproduction where we got reproductions from 1.98x to 2.0x but never greater than 2.0x. Our best lines were the bottom line of the box and then the middle average compared to the left line of the box. The variations in magnification could be a result of both the machining process as well as tracing process. While we were tracing the original image we may not have followed the lines exactly.

Lessons Learned:

One of the biggest lessons that we learnt is to prioritize having a good starting plan. During our creation process we had an original hole size that did not end up working so we had to go back and redrill our holes to fit the correct size. By doing this, we opened up a lot of room for error in

measurement through human or machine error. This could have been the reason we did not have a consistent 2.0x magnification. Another lesson we learnt was that it is not always best to be versatile/flexible. Our pen holder was made to allow for multiple different types of writing utensils. This was a nice thought to have but this also meant our measurements were not catered specifically to one type of writing utensil. If we had catered to one type of utensil this would have allowed us to increase our precision as we would have more specialized measurements.

Comparison of “as designed” to “as built”:

Our design between as built and as designed were very similar. We made sure to have our CAD model be very accurate to our final goal as we intended to use it as a blueprint for our final made design. There were not many differences between our CAD model and our physical, there may be some small differences in total bar length as we cut the bar length using the horizontal band saw as close as we could but I do not think we were able to get an exact measurement. This did not affect our overall magnification much as while using the mill, we were able to use the electronic measurement provided to get the correct distances between our holes.